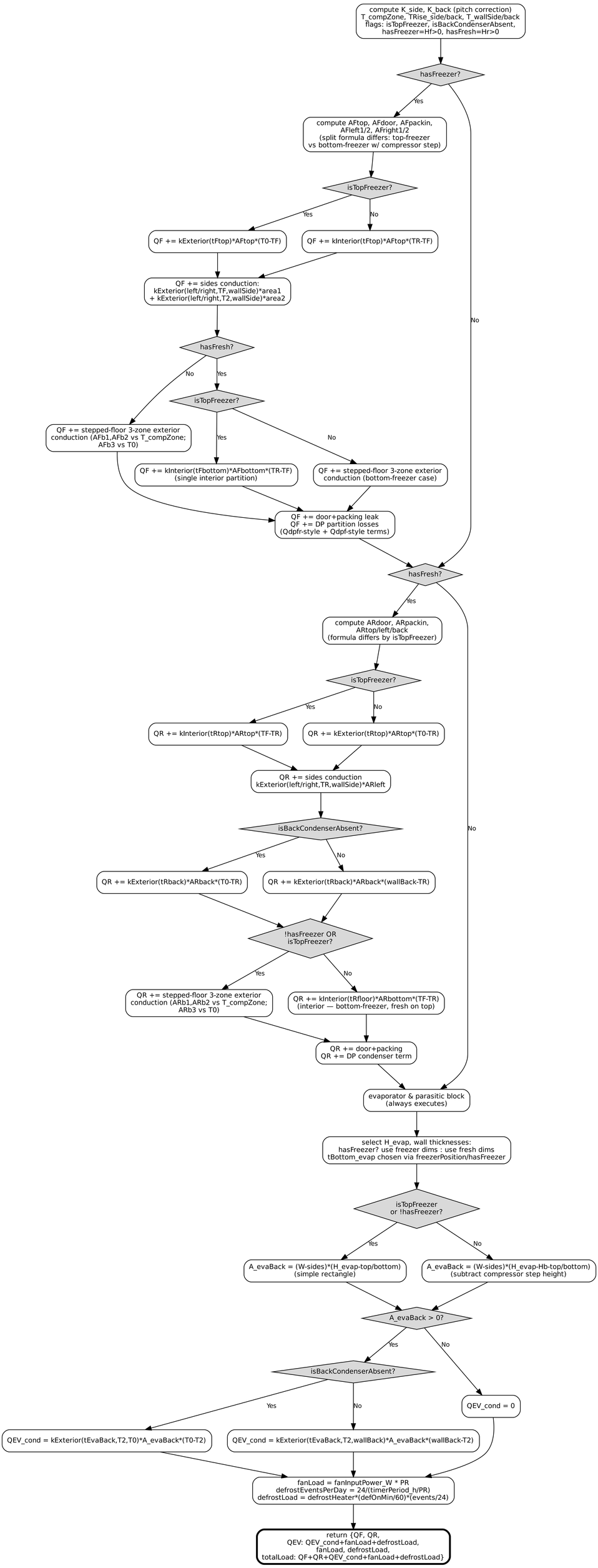
compute K_side, K_back (pitch correction)
T_compZone, TRise_side/back, T_wallSide/back
flags: isTopFreezer, isBackCondenserAbsent,
hasFreezer=Hf>0, hasFresh=Hr>0
hasFreezer?
Yes
No
compute AFtop, AFdoor, AFpackin,
AFleft1/2, AFright1/2
(split formula differs: top-freezer
vs bottom-freezer w/ compressor step)
isTopFreezer?
Yes
No
QF += kExterior(tFtop)*AFtop*(T0-TF)
QF += kInterior(tFtop)*AFtop*(TR-TF)
QF += sides conduction:
kExterior(left/right,TF,wallSide)*area1
+ kExterior(left/right,T2,wallSide)*area2
hasFresh?
No
Yes
isTopFreezer?
QF += stepped-floor 3-zone exterior
conduction (AFb1,AFb2 vs T_compZone;
AFb3 vs T0)
Yes
No
QF += kInterior(tFbottom)*AFbottom*(TR-TF)
(single interior partition)
QF += stepped-floor 3-zone exterior
conduction (bottom-freezer case)
QF += door+packing leak
QF += DP partition losses
(Qdpfr-style + Qdpf-style terms)
hasFresh?
Yes
No
compute ARdoor, ARpackin,
ARtop/left/back
(formula differs by isTopFreezer)
isTopFreezer?
Yes
No
QR += kInterior(tRtop)*ARtop*(TF-TR)
QR += kExterior(tRtop)*ARtop*(T0-TR)
QR += sides conduction
kExterior(left/right,TR,wallSide)*ARleft
isBackCondenserAbsent?
Yes
No
QR += kExterior(tRback)*ARback*(T0-TR)
QR += kExterior(tRback)*ARback*(wallBack-TR)
!hasFreezer OR
isTopFreezer?
Yes
No
QR += stepped-floor 3-zone exterior
conduction (ARb1,ARb2 vs T_compZone;
ARb3 vs T0)
QR += kInterior(tRfloor)*ARbottom*(TF-TR)
(interior — bottom-freezer, fresh on top)
QR += door+packing
QR += DP condenser term
evaporator & parasitic block
(always executes)
select H_evap, wall thicknesses:
hasFreezer? use freezer dims : use fresh dims
tBottom_evap chosen via freezerPosition/hasFreezer
isTopFreezer
or !hasFreezer?
Yes
No
A_evaBack = (W-sides)*(H_evap-top/bottom)
(simple rectangle)
A_evaBack = (W-sides)*(H_evap-Hb-top/bottom)
(subtract compressor step height)
A_evaBack > 0?
Yes
No
isBackCondenserAbsent?
QEV_cond = 0
Yes
No
QEV_cond = kExterior(tEvaBack,T2,T0)*A_evaBack*(T0-T2)
QEV_cond = kExterior(tEvaBack,T2,wallBack)*A_evaBack*(wallBack-T2)
fanLoad = fanInputPower_W * PR
defrostEventsPerDay = 24/(timerPeriod_h/PR)
defrostLoad = defrostHeater*(defOnMin/60)*(events/24)
return {QF, QR,
QEV: QEV_cond+fanLoad+defrostLoad,
fanLoad, defrostLoad,
totalLoad: QF+QR+QEV_cond+fanLoad+defrostLoad}